3-D Geometric Transformations

Computer Graphics, Unit 4 | B.Tech CSE, 3rd year | Classroom notes

Translation | Scaling | Rotation | Reflection | Shear | Composite transformations

Note for students: These notes cover the topics named in the unit title. How deep you go (for example, rotation about an arbitrary axis, reflection about an arbitrary plane) and the number of marks per topic depend on your own university syllabus, so always cross-check with it.

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1. Why we need 3-D transformations

A 3-D model (a car, a character, a building) is stored once, as a list of vertices in its own *local* coordinates. To place it in a scene, make it bigger, turn it, or mirror it, we do not rebuild the model. We apply a **geometric transformation** to every vertex.

> **Definition: Geometric transformation**
> A geometric transformation is a rule (a function) that maps every point **P(x, y, z)** of an object to a new point **P′(x′, y′, z′)**, changing the object's position, size, orientation or shape.

Real-world examples

- **Translation:** a game character walking forward; a camera moving down a street.
- **Rotation:** a door swinging on its hinge; a fan blade; a planet spinning.
- **Scaling:** pinch-to-zoom on a 3-D map; resizing a part in CAD.
- **Reflection:** building the left half of a car and mirroring it to get the right half.
- **Shear:** a building that leans in a simulation of a strong wind; italic-style slanting of 3-D text.
- **Composite:** a robot arm, where the hand moves because the shoulder, elbow and wrist all transform together.

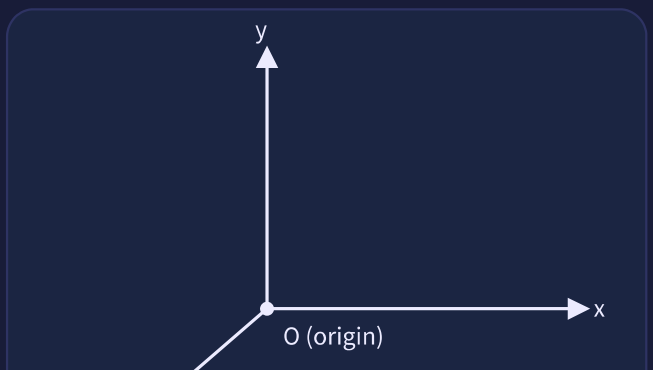
The big picture

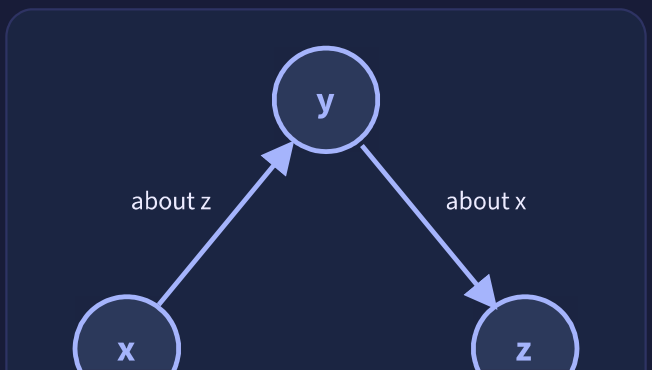
Transformation	What it changes	One-line idea
Translation	Position	Slide the object along a vector
Scaling	Size	Stretch or shrink along each axis
Rotation	Orientation	Turn the object about an axis
Reflection	Handedness (mirror image)	Flip the object across a plane
Shear	Shape	Slide layers sideways in proportion to height

2. Groundwork: axes and homogeneous coordinates

2.1 The 3-D coordinate system

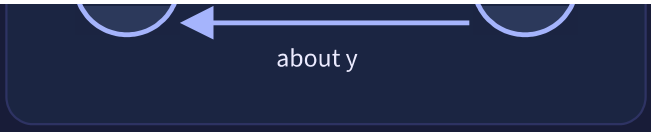
Most graphics textbooks use a **right-handed system**. Point the fingers of your right hand along +x and curl them toward +y. Your thumb then points along +z.







Right-handed axes: z points toward the viewer



Positive rotation moves x→y (about z), y→z (about x), z→x (about y)

Important: sign of the angle

A positive angle θ means **counter-clockwise rotation when you look from the positive end of the axis toward the origin** (right-hand rule: thumb along the axis, fingers curl in the positive direction). The cycle x→y→z→x in the figure is the easiest way to remember which way each axis turns.

2.2 Homogeneous coordinates

Definition: Homogeneous coordinates

A 3-D point (x, y, z) is written as a 4-element vector (x, y, z, 1). In general (x, y, z, w) represents the point (x/w, y/w, z/w). For ordinary points we keep **w = 1**.

Why do we need them? Scaling, rotation, reflection and shear are all of the form `P' = M·P` using a 3×3 matrix. Translation is `P' = P + t`, an *addition*, so it cannot be written with a 3×3 matrix. With an extra coordinate, **all five** become a single 4×4 matrix multiplication. That lets us multiply matrices together to build one combined matrix (composite transformation).

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b & c & t_x \\ d & e & f & t_y \\ g & h & i & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

The top-left 3×3 block (a to i) handles rotation, scaling, reflection and shear. The last column (t_x, t_y, t_z) handles translation. The bottom row stays [0 0 0 1] for all these **affine** transformations.

Convention used in these notes

We write points as **column vectors** and use **P′ = M·P**. Some books (and some exam papers) use row vectors, **P′ = P·M**, where every matrix is the *transpose* of the one shown here. Always follow the convention of your prescribed textbook.

3. Translation

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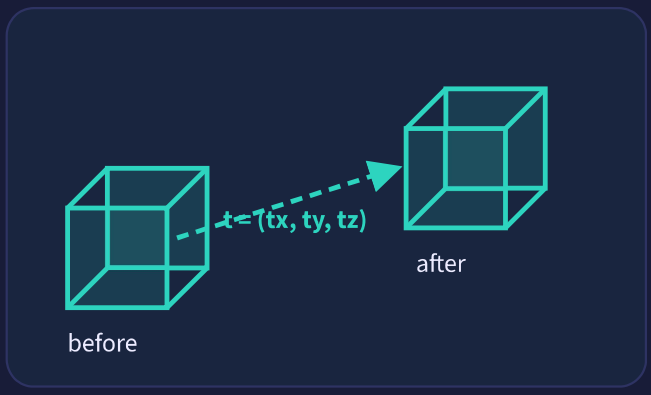
Definition: Translation

Translation moves every point of an object by the same distance in the same direction, defined by a translation vector (t_x, t_y, t_z). It is a **rigid-body** transformation: size, shape and orientation do not change.

Equations: $x' = x + t_x,\ \ y' = y + t_y,\ \ z' = z + t_z$

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Inverse: $T(t_x, t_y, t_z)^{-1} = T(-t_x, -t_y, -t_z)$ (move back).



Every vertex moves by the same vector, so the cube is unchanged except for its position

Worked example 1

Translate P(2, 3, 4) by t = (3, −1, 2).

$x' = 2 + 3 = 5$, $y' = 3 - 1 = 2$, $z' = 4 + 2 = 6$. **P′ = (5, 2, 6).**

Check with the matrix: the last column adds (3, −1, 2) to (2, 3, 4), giving the same answer.

4. Scaling

Definition: Scaling

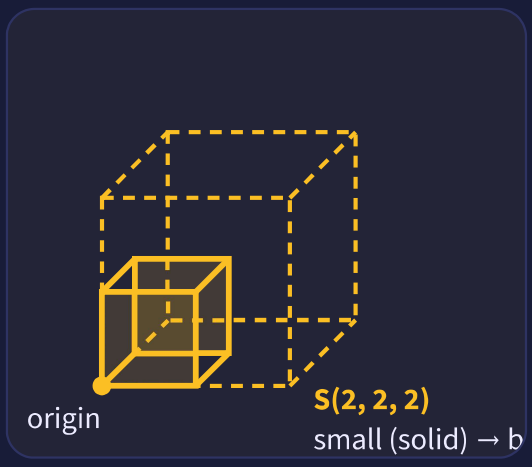
Scaling changes the size of an object by multiplying each coordinate by a scale factor: s_x, s_y, s_z. If $s_x = s_y = s_z$ it is **uniform scaling** (shape preserved). Otherwise it is **non-uniform (differential) scaling** (object is stretched).

Equations (about the origin): $x' = s_x \cdot x,\quad y' = s_y \cdot y,\quad z' = s_z \cdot z$

$$\begin{pmatrix} x' \\ y' \\ z' \\ 1 \end{pmatrix} = \begin{pmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Inverse: $S(s_x, s_y, s_z)^{-1} = S(1/s_x, 1/s_y, 1/s_z)$. Scale factors must not be 0.

- $s > 1$: enlarges. $0 < s < 1$: shrinks. $s < 0$: also reflects (see Section 6).
- Scaling about the origin also moves the object *away from* (or toward) the origin, because positions are multiplied too.



Uniform scaling about the origin. The corner at the origin stays fixed.

Scaling about a fixed point (x_f, y_f, z_f)

To scale an object about a point of our choice (say, its centre) so that the point does not move:

1. Translate the fixed point to the origin: $T(-x_f, -y_f, -z_f)$
2. Scale: $S(s_x, s_y, s_z)$
3. Translate back: $T(x_f, y_f, z_f)$

$M = T(x_f, y_f, z_f) \cdot S(s_x, s_y, s_z) \cdot T(-x_f, -y_f, -z_f)$

$$\begin{pmatrix} s_x & 0 & 0 & (1-s_x)x_f \\ 0 & s_y & 0 & (1-s_y)y_f \\ 0 & 0 & s_z & (1-s_z)z_f \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Worked example 2: scaling about the origin

Scale P(1, 2, 3) with S(2, 3, 0.5): P′ = (2·1, 3·2, 0.5·3) = **(2, 6, 1.5)**.

Worked example 3: fixed-point scaling

Scale P(3, 2, 1) by 2 (uniform) about the fixed point (1, 1, 1).

Step 1, translate by (−1, −1, −1): (2, 1, 0). Step 2, scale by 2: (4, 2, 0). Step 3, translate back by (1, 1, 1): **(5, 3, 1)**.

Shortcut check: $P' = P_f + s(P - P_f) = (1,1,1) + 2\cdot(2,1,0) = (5,3,1)$. ✓

Real-world use

Enlarging a 3-D model about its own centre, so it grows in place instead of drifting across the screen.

5. Rotation

Definition: 3-D rotation

Rotation turns an object through an angle θ about an **axis of rotation**. In 2-D we rotate about a point; in 3-D we must choose a **line** (axis). Rotation is a rigid-body transformation (no change in size or shape).

When the axis is a coordinate axis, the coordinate along that axis does not change, and the other two coordinates rotate exactly like a 2-D rotation.

5.1 Rotation about the z-axis

$x' = x\cos\theta - y\sin\theta$, $y' = x\sin\theta + y\cos\theta$, $\mathbf{z' = z}$

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{R_z(\theta)}$$

5.2 Rotation about the x-axis

$\mathbf{x' = x}$, $y' = y\cos\theta - z\sin\theta$, $z' = y\sin\theta + z\cos\theta$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{R_x(\theta)}$$

5.3 Rotation about the y-axis

$x' = x\cos\theta + z\sin\theta$, $\mathbf{y' = y}$, $z' = -x\sin\theta + z\cos\theta$

$$\begin{pmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{R_y(\theta)}$$

Important: why does R_y look "upside down"?

Follow the cycle x→y→z→x. For R_z the pair is (x, y), for R_x it is (y, z), and for R_y it is (z, x). In R_y the pair is (z, x), not (x, z), so the sign of the sin θ terms is swapped compared with the pattern of the other two. This is not a typing error. It is a very common exam slip.

5.4 Properties

- **Inverse:** $R(\theta)^{-1} = R(-\theta) = R(\theta)^T$ (rotation matrices are **orthogonal**).
- **Determinant = +1**, so orientation (handedness) is kept.
- Rotations about the *same* axis commute. Rotations about *different* axes **do not** commute: $R_x(90°)\cdot R_y(90°) \neq R_y(90°)\cdot R_x(90°)$.

Worked example 4: quick checks (θ = 90°, cos θ = 0, sin θ = 1)

- $R_z(90°)$ on (1, 0, 0): $x' = 0, y' = 1, z' = 0$ → **(0, 1, 0)** (x-axis turns into y-axis).
- $R_x(90°)$ on (0, 2, 0): $y' = 0, z' = 2$ → **(0, 0, 2)** (y turns into z).
- $R_y(90°)$ on (0, 0, 1): $x' = 1, z' = 0$ → **(1, 0, 0)** (z turns into x). Also $R_y(90°)$ on (2, 0, 0) gives (0, 0, −2).

These match the cycle x→y→z→x. If your answer disagrees, recheck the matrix.

5.5 Rotation about an axis parallel to a coordinate axis

Rotate about a line through a pivot point (x_p, y_p, z_p) parallel to, say, the z-axis:

$$\mathbf{M = T(x_p, y_p, z_p) \cdot R_z(\theta) \cdot T(-x_p, -y_p, -z_p)}$$

5.6 Rotation about an arbitrary axis (outline)

Suppose the axis passes through point P_1 with unit direction vector (a, b, c). Let $d = \sqrt{(b^2 + c^2)}$ (assume $d \neq 0$; if $d = 0$ the axis is already the x-axis).

1. Translate P_1 to the origin
2. $R_x(\alpha)$: put the axis in the xz-plane, with $\cos\alpha = c/d$, $\sin\alpha = b/d$
3. $R_y(\beta)$: align the axis with +z, with $\cos\beta = d$, $\sin\beta = -a$
4. $R_z(\theta)$: the required rotation
5. Undo step 3: $R_y(-\beta)$
6. Undo step 2: $R_x(-\alpha)$
7. Translate back

$$M = T(P_1) \cdot R_x(-\alpha) \cdot R_y(-\beta) \cdot R_z(\theta) \cdot R_y(\beta) \cdot R_x(\alpha) \cdot T(-P_1)$$

Sign convention

The signs of sin α and sin β above go with the R_x and R_y matrices written in this section. Some books define the intermediate angles differently, so signs can change. Use your textbook's version in the exam, and verify by checking that the axis ends up on the z-axis.

Real-world examples

A door swinging on a vertical hinge (rotation about a line parallel to the y-axis through the hinge). The Earth spinning about its tilted axis. A drone rolling, pitching and yawing.

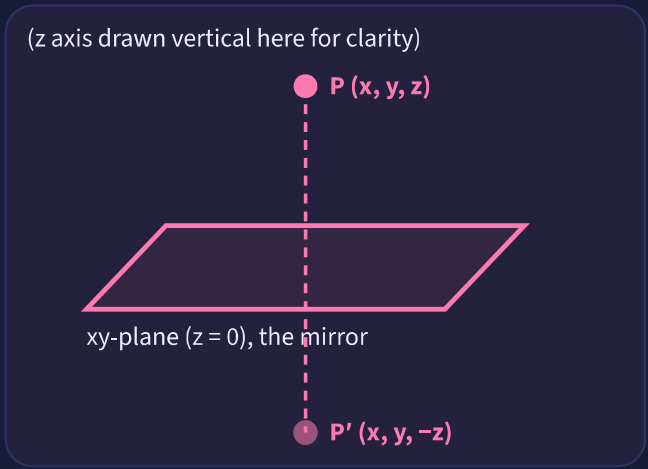
6. Reflection

Definition: Reflection

Reflection produces the **mirror image** of an object. In 3-D the mirror is a **plane**. Each point is moved to the opposite side of the plane, at the same perpendicular distance.

Reflection about	Effect on coordinates	Matrix (4×4)
xy-plane (z = 0)	$z' = -z$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

yz-plane (x = 0)	$x' = -x$	$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$
xz-plane (y = 0)	$y' = -y$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$
Origin (point reflection)	$x' = -x, y' = -y, z' = -z$	$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$



Reflection about the xy-plane flips only the z coordinate

- **Inverse:** reflecting twice gives the original, so a reflection matrix is its own inverse.
- **Determinant = −1.** A reflection turns a right-handed object into a left-handed one. No rotation can do that.
- Reflection about an arbitrary plane: rotate/translate so that the plane becomes a coordinate plane, reflect, then undo (the same "move, do it, move back" idea).

Worked example 5

Reflect P(2, 3, 4) about the xz-plane: only y changes sign → **P′ = (2, −3, 4)**. About the xy-plane it would be (2, 3, −4). Through the origin: (−2, −3, −4).

Real-world examples

Modelling only one half of a symmetric car, aircraft or face and mirroring it. Drawing the reflection of a building on a lake surface (mirror plane = water level).

7. Shear

Definition: Shear

Shear distorts the shape of an object by shifting one coordinate in proportion to another coordinate, like sliding the pages of a deck of cards. Parallel planes slide past one another while the object stays attached to one reference plane.

Z-axis shear (reference plane z = 0)

$x' = x + a \cdot z$, $y' = y + b \cdot z$, $\mathbf{z' = z}$

$$\begin{pmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{SH_z(a, b)}$$

X-axis shear (reference plane x = 0)

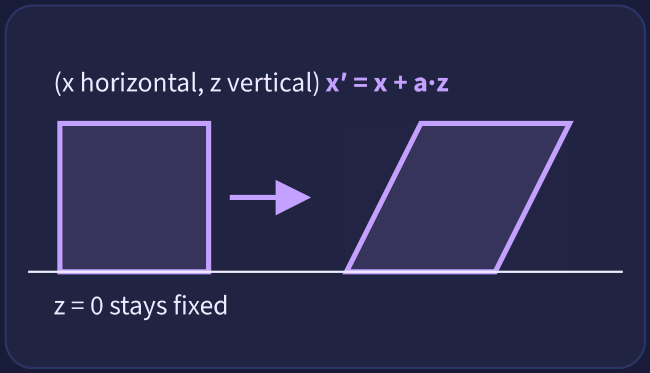
$\mathbf{x' = x}$, $y' = y + b \cdot x$, $z' = z + c \cdot x$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ b & 1 & 0 & 0 \\ c & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \mathbf{SH_x(b, c)}$$

Y-axis shear (reference plane y = 0)

$x' = x + a \cdot y$, $\mathbf{y' = y}$, $z' = z + c \cdot y$

$$\begin{pmatrix} 1 & a & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & c & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = SH_y(a, c)$$



Z-shear in the xz view: the top layer slides right in proportion to its height

- **Determinant = 1**, so volume is preserved. Lengths and angles are **not** preserved. Parallel lines stay parallel.
- **Inverse:** shear by the negative amounts, e.g. $SH_z(-a, -b)$.

Worked example 6

Apply z-axis shear with a = 1, b = 2 to P(1, 2, 3): x′ = 1 + 1·3 = 4, y′ = 2 + 2·3 = 8, z′ = 3 → **P′ = (4, 8, 3)**. A point with z = 0 would not move at all.

Real-world examples

A tall building leaning sideways in a wind animation. Slanted (italic-like) 3-D lettering. A stack of papers pushed sideways. Deforming a character's body in simple animation.

8. Comparison tables

8.1 Matrix and key facts

Transformation	Matrix idea	Inverse	Determinant of 3×3 part
Translation	Identity with t in last column	T(−t)	1
Scaling	Diagonal (s_x, s_y, s_z)	S($1/s_x, 1/s_y, 1/s_z$)	$s_x \cdot s_y \cdot s_z$

Rotation	cos θ, sin θ block	$R(-\theta) = R^T$	+1
Reflection	Diagonal with one (or three) −1	Itself	−1 (plane) / −1 (origin)
Shear	Identity with extra off-diagonal terms	Shear by negative amounts	1

8.2 What is preserved?

Transformation	Lengths	Angles	Parallel lines	Volume	Handedness
Translation	Yes	Yes	Yes	Yes	Kept
Rotation	Yes	Yes	Yes	Yes	Kept
Reflection	Yes	Yes	Yes	Yes (size)	**Flipped**
Uniform scaling	No (× s)	Yes	Yes	No (× s^3)	Kept (s > 0)
Non-uniform scaling	No	No (in general)	Yes	No	Kept (all s > 0)
Shear	No	No	Yes	Yes	Kept

> **Key idea**
> All five are **affine transformations**: straight lines stay straight and parallel lines stay parallel. Translation, rotation and reflection are also called **rigid** (isometric) because distances are preserved.

8.3 2-D versus 3-D

Point	2-D	3-D
Point representation	(x, y, 1)	(x, y, z, 1)
Matrix size	3×3	4×4
Rotation	About a point (one angle)	About an axis (three basic matrices)
Reflection	About a line	About a plane
Rotations commute?	Yes (about the same point)	No (different axes)
Shear	Along x or y	Along x, y or z (two coordinates shift)

8.4 Row-vector versus column-vector convention

Item	Column vector (these notes)	Row vector
Formula	$P' = M \cdot P$	$P' = P \cdot M$
Translation in matrix	Last column	Last row
Combining M_1 then M_2	$M = M_2 \cdot M_1$ (right to left)	$M = M_1 \cdot M_2$ (left to right)
Matrices	As shown here	Transpose of those shown here

9. Composite transformations

Definition: Composite (concatenated) transformation

A composite transformation is a sequence of basic transformations combined into **one matrix** by multiplying their matrices. Applying that single matrix to every vertex is faster than applying each step separately.

9.1 The rule

If we apply M_1 first, then M_2, then M_3 (column-vector convention):

$$\mathbf{M = M_3 \cdot M_2 \cdot M_1} \quad \text{and} \quad \mathbf{P' = M \cdot P}$$

Important points

- **Read right to left.** The matrix closest to P acts first.
- Matrix multiplication is **associative** (you may group it as you like) but **not commutative** in general (the order matters).
- Order can be swapped only in special cases: translation with translation, scaling with scaling, rotations about the *same* axis, and uniform scaling with any rotation.

Real-world analogy

"Put on your socks, then your shoes" is not the same as "shoes, then socks." Likewise, "move 2 units and then rotate" is not the same as "rotate and then move 2 units."

9.2 Order matters: proof by example

Worked example 7

Take the point P(0, 0, 0). Let T = translate by (2, 0, 0) and R = $R_z(90°)$.

T first, then R: (0,0,0) → (2,0,0) → R gives (0, 2, 0).

R first, then T: (0,0,0) → (0,0,0) → T gives (2, 0, 0).

The answers differ, so **R·T ≠ T·R**.

9.3 Typical composite patterns

Goal	Composite matrix (apply right to left)
Scale about a fixed point F	$T(F) \cdot S \cdot T(-F)$
Rotate about an axis parallel to z through pivot P	$T(P) \cdot R_z(\theta) \cdot T(-P)$
Rotate about an arbitrary axis	$T(P_1) \cdot R_x(-\alpha) \cdot R_y(-\beta) \cdot R_z(\theta) \cdot R_y(\beta) \cdot R_x(\alpha) \cdot T(-P_1)$
Reflect about a plane that does not pass through the origin (parallel to a coordinate plane)	T(d) · Reflect · T(−d), where d is a point on the plane
Standard scene placement of a model (common practice in graphics)	Translate · Rotate · Scale (scale first, then rotate, then move)

9.4 Practical examples

Worked example 8: building the composite matrix

Scale uniformly by 2, then translate by (1, 2, 3). Here M = T · S:

$$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Test with P(1, 1, 1): scale gives (2, 2, 2), translate gives (3, 4, 5). Using M: (2+1, 2+2, 2+3) = (3, 4, 5). ✓

Worked example 9: rotation about a parallel axis

Rotate P(3, 1, 5) by 90° about the line through (1, 1, 0) parallel to the z-axis.

1. Translate pivot to origin, T(−1, −1, 0): (2, 0, 5)

1. Translate pivot to origin, T(−1, −1, 0): (2, 0, 5).
2. $R_z(90°)$: (x, y) → (−y, x), giving (0, 2, 5).
3. Translate back, T(1, 1, 0): **(1, 3, 5)**.

Check: the z value (5) is unchanged, and the distance from the axis is still 2. ✓

Worked example 10: scale a cube about its own centre

A cube has opposite corners (1, 1, 1) and (3, 3, 3), so its centre is (2, 2, 2). Double its size without moving the centre: M = T(2,2,2) · S(2,2,2) · T(−2,−2,−2).

Corner (1,1,1) → (−1,−1,−1) → (−2,−2,−2) → **(0, 0, 0)**. Corner (3,3,3) → (1,1,1) → (2,2,2) → **(4, 4, 4)**.

The new cube runs from (0,0,0) to (4,4,4), still centred at (2,2,2).

Real-world example: a robot arm or the solar system

Each child part (forearm, hand, moon) has its own transformation, and the final matrix is the product of its parent's matrices. A moon's matrix = (Sun-to-Earth placement) · (Earth's spin/orbit) · (Moon's own orbit). This chaining of matrices is the idea behind **scene graphs** and matrix stacks.

10. Common student mistakes

Mistake	Why it is wrong	How to avoid it
Using a 3×3 matrix for translation	Translation is addition, which a 3×3 matrix cannot do	Use 4×4 and (x, y, z, 1)
Multiplying composite matrices left to right	With column vectors, the first transformation must be the right-most matrix	Write the steps in a list, then reverse them to build M
Assuming rotations commute	Rotations about different axes give different results depending on order	Test with 90° rotations of a book
Wrong signs in R_y	The pair is (z, x), so the sin θ signs are swapped	Verify with $R_y(90°)$ on (0,0,1) → (1,0,0)
Rotating about a pivot without translating	R_z, R_x, R_y turn the object about the *origin*	Always use T(P) · R · T(−P)
Scaling about the origin and expecting the object to stay in place	Positions are multiplied too, so the object moves	Use fixed-point scaling

Degrees versus radians	Calculators and code (sin, cos) often expect radians	Convert: rad = deg × π/180
Mixing row and column conventions	Matrices become transposed and the order reverses	Decide the convention first, and state it in the answer
Forgetting w = 1 or the bottom row [0 0 0 1]	Without it, the matrix is not affine, and the result is wrong	Check the last row before multiplying
Thinking shear and rotation are the same	Rotation keeps lengths; shear does not	Compare the lengths of a side before and after

11. Interview questions

Tap a question to see a model answer.

1. Why do we use homogeneous coordinates in graphics?

2. Why is a 4×4 matrix used for 3-D transformations?

3. Are 3-D rotations commutative?

4. What is the inverse of a rotation matrix?

5. What does a determinant of −1 mean for a transformation?

6. Difference between uniform and non-uniform scaling?

7. How would you rotate an object about its own centre?

8. How do you rotate about an arbitrary axis?

9. What is an affine transformation?

10. What is gimbal lock?

11. In OpenGL, in what order are transformations applied?

12. Is shear a rigid-body transformation?

12. University examination questions

> **About the marks**
> Marks are indicative only. Match them to your own university's paper pattern.

Short answer

1. Define 3-D translation and write its transformation matrix. **2**
2. Why are homogeneous coordinates used? **2**
3. Write the matrix for reflection about the xy-plane. **2**
4. What is the inverse of the scaling matrix $S(s_x, s_y, s_z)$? **2**
5. Write the matrix for a z-axis shear. **2**
6. Differentiate between uniform and differential scaling. **2**
7. State any two properties of composite transformations. **2**

Medium answer

8. Derive the rotation matrix about the z-axis and write the matrices for the x- and y-axes. **5**
9. A point P(4, 2, 1) is translated by (−1, 3, 2) and then scaled by (2, 2, 2) about the origin. Find the final point and the composite matrix. **5**
10. Explain scaling about a fixed point with a matrix and a numerical example. **5**
11. Write brief notes on 3-D reflection and shear with matrices. **5**
12. Show that translation followed by rotation is not the same as rotation followed by translation. **5**
13. Find the new coordinates of P(1, 2, 3) after a rotation of 90° about the x-axis. **3**

▶ **Answers to the numerical questions (9 and 13)**

13. Five long-answer questions

1. **Basic transformations.** Explain the concept of homogeneous coordinates. Derive the 4×4 matrices for 3-D translation, scaling and rotation about each of the three coordinate axes. State the inverse of each. Apply them to a point of your choice. **10**

2. **Rotation about an arbitrary axis.** Explain the steps required to rotate an object about an arbitrary axis passing through a given point, with a neat diagram and the final composite expression. Solve a numerical example for an axis parallel to a coordinate axis. 10
3. **Composite transformations.** What is a composite transformation? Explain why the order of matrices matters, with an example. Find the composite matrix to scale a cube with corners (1,1,1) and (3,3,3) to twice its size about its own centre, and list the new corners. 10
4. **Reflection and shear.** Derive the matrices for reflection about the three coordinate planes and for shear along the x-, y- and z-axes. Compare their effects on lengths, angles, volume and handedness. Give a real-life application of each. 10
5. **Properties and comparison.** Compare translation, rotation, scaling, reflection and shear in a table using: effect, matrix form, inverse, determinant, and what is preserved. Explain which of them are rigid-body and which are affine transformations. 10

14. 5-minute classroom activity: "Does the order matter?"

Format: pairs. One pen, one sheet each. No calculator needed (use 90° so cos = 0, sin = 1).

Time	What happens
0:00 – 0:30	The teacher writes the task on the board and explains the right-to-left rule.
0:30 – 2:30	**Partner A** works out Plan 1. **Partner B** works out Plan 2 (same point, two orders).
2:30 – 3:30	Partners swap sheets and check each other's arithmetic.
3:30 – 4:30	Pairs discuss: *Are the final points the same? Why or why not?*
4:30 – 5:00	The teacher asks two or three pairs to share their results, then reveals the answer.

Task

Start with P(1, 2, 3) and use these three transformations: **T** = translate by (2, 0, 0), **R** = $R_z(90°)$, **F** = reflect about the xy-plane.

Plan 1: T, then R, then F. **Plan 2:** R, then T, then F.

Then write the composite matrix expression for each plan.

Instructor answer key

Plan 1: (1,2,3) → T: (3,2,3) → $R_z(90°)$, (x,y)→(−y,x): (−2,3,3) → F: **(−2, 3, −3)**. $M_1 = F \cdot R \cdot T$

Plan 2: (1,2,3) → R: (−2,1,3) → T: (0,1,3) → F: **(0, 1, −3)**. $M_2 = F \cdot T \cdot R$

> The results differ, so the **order matters**. Discussion point: why would swapping the order of R and F *not* change anything here? (Because R_z only affects x and y, and F only affects z, so those two commute.)

15. Recap and formula sheet

Short recap

- 3-D transformations are done with **4×4 matrices** on points written as (x, y, z, 1).
- **Translation** adds t; **scaling** multiplies by s; **rotation** turns about an axis; **reflection** mirrors in a plane; **shear** slides layers sideways.
- Rotations about different axes **do not commute**. Rotation matrices have **inverse = transpose**.
- Reflection has **determinant −1** (flips handedness); shear has determinant 1 (volume kept).
- For a pivot/fixed point: **move to origin → transform → move back**.
- Composite: $\mathbf{M = M_n \cdots M_2 \cdot M_1}$. The first step is on the right.

Formula sheet

Operation	Equations
Translation	$x' = x + t_x,\ y' = y + t_y,\ z' = z + t_z$
Scaling	$x' = s_x x,\ y' = s_y y,\ z' = s_z z$
Rotation about z	$x' = x\cos\theta - y\sin\theta,\ y' = x\sin\theta + y\cos\theta,\ z' = z$
Rotation about x	$y' = y\cos\theta - z\sin\theta,\ z' = y\sin\theta + z\cos\theta,\ x' = x$
Rotation about y	$z' = z\cos\theta - x\sin\theta,\ x' = x\cos\theta + z\sin\theta,\ y' = y$
Reflection (xy / yz / xz)	$z' = -z$ / $x' = -x$ / $y' = -y$
Shear (z-axis)	$x' = x + a\,z,\ y' = y + b\,z,\ z' = z$
Fixed-point scaling	$T(F) \cdot S \cdot T(-F)$
Rotation about a parallel axis	$T(P) \cdot R \cdot T(-P)$